```
PS C:\Users\arlet\OneDrive\Documentos\ManejoMasivo\Segundo_P\examen_2p> Get-CimInstance Win32_Process -Filter "name = 'python.exe'" | Select-Object ProcessId
,CommandLine
PS C:\Users\arlet\OneDrive\Documentos\ManejoMasivo\Segundo_P\examen_2p> python -u .\generar_dataset.py 125054136
Generando dataset para 125054136... (tarda unos minutos)
Listo
```

```
PS C:\Users\arlet\OneDrive\Documentos\ManejoMasivo\Segundo_P\examen_2p> docker compose ps
NAME                IMAGE                  COMMAND                  SERVICE   CREATED          STATUS                    PORTS
examen_2p-api-1     examen_2p-api          "python -u app.py"       api       45 seconds ago   Up 38 seconds             0.0.0.0:5051->5000/tcp, [::]:505
->5000/tcp
examen_2p-db-1      postgres:16-alpine     "docker-entrypoint.s…"   db        48 seconds ago   Up 45 seconds (healthy)   5432/tcp
examen_2p-hadoop-1  examen-p2-hadoop:3.3.6  "/usr/local/bin/entr…"   hadoop    48 seconds ago   Up 44 seconds (healthy)   22/tcp
PS C:\Users\arlet\OneDrive\Documentos\ManejoMasivo\Segundo_P\examen_2p>
```

```
PS C:\Users\arlet\OneDrive\Documentos\ManejoMasivo\Segundo_P\examen_2p> docker compose exec hadoop hadoop version
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Compiled with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /opt/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar
```

```
PS C:\Users\arlet\OneDrive\Documentos\ManejoMasivo\Segundo_P\examen_2p> docker compose exec -u alumno hadoop ssh localhost hostname
hadoop
PS C:\Users\arlet\OneDrive\Documentos\ManejoMasivo\Segundo_P\examen_2p> docker compose exec api ssh -i /ssh/id_ed25519 -o BatchMode=yes -o StrictHostKeyCheck
ing=accept-new alumno@hadoop hostname
Warning: Permanently added 'hadoop' (ED25519) to the list of known hosts.
hadoop
```

```
TIEMPO HADOOP STREAMING: 99.956 segundos
RESULTADO (TOP 10):
FAR-0007        131431409.33
FAR-0019        129340829.53
FAR-0006        128703932.54
FAR-0023        128545889.56
FAR-0010        128422499.44
FAR-0008        127756838.96
FAR-0011        127716765.88
FAR-0027        127691255.59
FAR-0018        127687769.55
FAR-0032        127623381.72
```

```
        FILE: Number of write operations=0
Map-Reduce Framework
        Map input records=5000001
        Map output records=5000000
        Map output bytes=81088804
        Map output materialized bytes=91088

        Input split bytes=979
        Combine input records=0
        Combine output records=0
        Reduce input groups=602
        Reduce shuffle bytes=91088870
        Reduce input records=5000000
        Reduce output records=10
        Spilled Records=10000000
        Shuffled Maps =11
        Failed Shuffles=0
        Merged Map outputs=11
```

```
FAR-0032        12762python .\mapreduce\secuencial.py .\datos\ventas_125054136.csv .\salidas\secuencial.tsvdo_P\examen_2p>
Tiempo Python secuencial: 38.496 segundos
Registros válidos: 5000000
Resultado: salidas\secuencial.tsv
PS C:\Users\arlet\OneDrive\Documentos\ManejoMasivo\Segundo_P\examen_2p> python .\scripts\comparar.py
>> python .\scripts\actualizar_readme.py
COINCIDEN EXACTAMENTE: los 10 productos, el orden y los ingresos.
Tabla de rendimiento actualizada en README.md.
```

```
PS C:\Users\arlet\OneDrive\Documentos\ManejoMasivo\Segundo_P\examen_2p> docker compose exec api python /app/scripts/cargar_resultados.py
>>
>> Invoke-RestMethod http://localhost:5051/resultados
>>
>> Invoke-RestMethod http://localhost:5051/resultados/FAR-0007
Cargados 10 productos en PostgreSQL (db).

ingreso_total producto
------------- --------
131431409.33  FAR-0007
129340829.53  FAR-0019
128703932.54  FAR-0006
128545889.56  FAR-0023
128422499.44  FAR-0010
127756838.96  FAR-0008
127716765.88  FAR-0011
127691255.59  FAR-0027
127687769.55  FAR-0018
127623381.72  FAR-0032
131431409.33  FAR-0007
```

```
>> Invoke-RestMethod http://localhost:5051/resultados/PRUEBA-0001

ingreso_total producto
------------- --------
100.50        PRUEBA-0001
100.50        PRUEBA-0001
```

```
PS C:\Users\arlet\OneDrive\Documentos\ManejoMasivo\Segundo_P\examen_2p> docker compose down
>> docker compose up -d
>> Invoke-RestMethod http://localhost:5051/resultados/FAR-0007
[+] down 4/4
 ✓ Container examen_2p-api-1      Removed      3.5s
 ✓ Container examen_2p-db-1       Removed      0.5s
 ✓ Container examen_2p-hadoop-1   Removed      0.5s
 ✓ Network examen_2p_datamart_red Removed      0.3s
[+] up 4/4
 ✓ Network examen_2p_datamart_red Created      0.1s
 ✓ Container examen_2p-db-1       Healthy      6.8s
 ✓ Container examen_2p-hadoop-1   Healthy      6.8s
 ✓ Container examen_2p-api-1      Started      7.1s

ingreso_total producto
------------- --------
131431409.33  FAR-0007
```